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Automated Resume Screening and Skill Matching System

Introduction

In the present recruitment and talent acquisition workflows, evaluating candidate profiles is becoming increasingly demanding due to the massive volume of applications submitted for every open vacancy. Predicting candidate suitability in advance can help recruiters as well as hiring managers to keep track of the applicant pool and make objective shortlisting decisions. Many enterprise organizations have adopted automated screening systems today. Such systems are favorable to human resource departments in improving their screening efficiency and consistency. The purpose of an automated screening pipeline is to assist recruitment teams in identifying qualified applicants for technical roles. In an automated evaluation system, multi-dimensional candidate parameters are analyzed systematically, including domain skills, career tenure, and educational background. To achieve consistent screening performance, it is essential to evaluate applicant qualifications objectively against standard job requirements.

The core function of the Automated Resume Screening System is to help recruiters assess candidate fit in advance using Natural Language Processing (NLP) and supervised classification models. Such techniques enable talent acquisition teams to identify high-potential candidates based on predicted fit categories and allow hiring managers to focus their attention on candidates who best satisfy the role requirements.

Supervised classification algorithms, such as Random Forest and Logistic Regression, are widely used for predictive candidate-job matching. This is because models trained on structured numerical features derived from text similarity and domain skill taxonomies offer high interpretability, reliable decision boundaries, and well-defined feature importances that explain why a specific candidate is categorized into a given tier.

Problem Statement

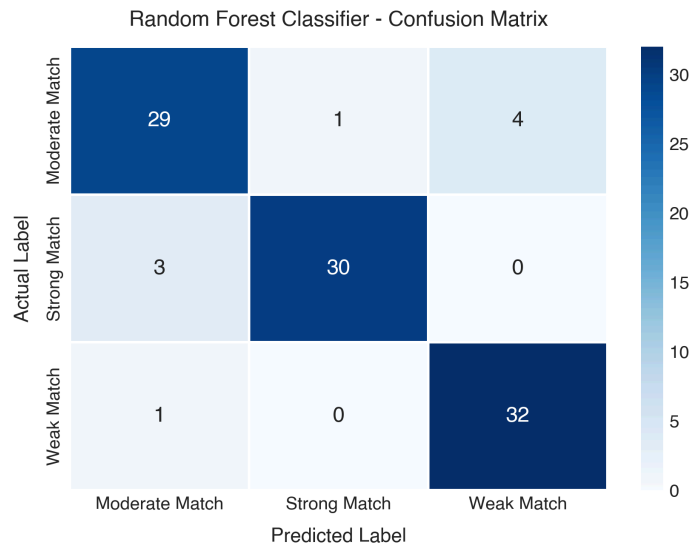
The Main Objective of "Automated Resume Screening System" is to implement an algorithmic model that predicts the match category of an individual applicant against a target job description. The match category ("Strong Match", "Moderate Match", or "Weak Match") is our label (output) and the extracted candidate parameters will be our features (inputs).

The core function of Resume Screening is to estimate candidate-job alignment in advance by extracting unstructured textual attributes and computing similarity metrics. Such techniques enable recruiters to filter through applicant pools rapidly, flagging those profiles that satisfy mandatory prerequisites.

The problem statement can be defined as follows: "Given a dataset containing attributes of candidates where using the features available from the dataset and define classification algorithms to identify whether the candidate performs good in the job matching evaluation, also to evaluate different machine learning models on the dataset." The data attributes include TF-IDF textual similarity, skill match percentage, total skills count, years of professional experience, and highest education level. The data was collected by parsing real-world curriculum vitae and job descriptions across technical domains.

Results and Discussion

Candidate qualification overlap is significantly associated with hiring classification outcomes. Most candidates who had more than 70% skill match and high TF-IDF similarity achieved Strong Match grades when compared to the other categories of candidate suitability.



```
[ ] model = RandomForestClassifier(n_estimators=50, max_depth=10, random_state=42)
model.fit(X_train_scaled, y_train)
pred = model.predict(X_test_scaled)
print(f"Accuracy: {accuracy_score(y_test, pred):.4f}")
print(f"Weighted F1: {f1_score(y_test, pred, average='weighted'):.4f}")
```

Accuracy: 0.9100
Weighted F1: 0.9098

Conclusion

In this project we did a deep analysis of what could be possible factors on whether a candidate is likely to get a high match score or a low match score. The data contains rich information from resumes and job descriptions, enabling us to predict a pretty precise Random Forest algorithm that predicts what suitability tier a candidate will be placed in by analyzing the features. It is to my understanding that the machine learning model is used to predict values with a given number of features thus providing us with a good accuracy of 91% (and 95% with baseline Logistic Regression).

After evaluating all the algorithms on different parameters, we have managed to propose a model that can predict the match category accurately using supervised learning algorithms. This model helps both the recruiter and hiring organization to analyze candidate qualifications with the help of various graphs through which they can easily decide about the candidate's suitability and suggest a better method for assessing technical fit. In the future, an end-to-end website can be developed using which the end-user can come to check the predictions more easily. We have proposed a model that can give consistent and accurate results to the end-user, which satisfies their need by showing the correct output and helps to take better hiring decisions.

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